Unconfounded but inflated causal estimates

Mathematical description

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I Setting

I. 1 Objectives

The objective of our article is to compare "biases" resulting from exaggeration in causal identification strategies on the one hand and from confounders in "naive" regression analysis on the other hand. This document aims to provide a mathematical derivation to illustrate this trade-off. Such a derivation may provide some intuition regarding the various mechanisms at play. It does not aim in any way to provide a general proof but merely a specific one, tied to the setting considered. Following a framework common to most analyses similar to this present one, while we keep the setting as general as possible, we have to make certain assumptions, in particular regarding the data generating process considered.

The objective in this mathematical derivation is to compare the exaggeration ratio for a naive regression with omitted variable bias (OVB) to that of an Instrumental Variable (IV) model in which we instrument the variable affected by the omitted variable and of a reduced form (RED) estimation in which we directly regress the outcome on the instrument. We expect the naive regression to be biased but more precise than the IV and the reduced form and therefore less subject to exaggeration problems. Roughly, we aim to derive the conditions under which the exaggeration due to the loss in precision is greater than the omitted variable bias. The objective is to compare the following quantities:

$$\mathbb{E}\left[\hat{\beta}_{\text{OVB}} |\ \beta_1, \sigma_{\text{OVB}}, \hat{\beta}_{\text{OVB}} > z_\alpha \sigma_{\text{OVB}}\right]$$
$$\mathbb{E}\left[\hat{\beta}_{\text{IV}} |\ \beta_1, \sigma_{\text{IV}}, \hat{\beta}_{\text{IV}} > z_\alpha \sigma_{\text{IV}}\right]$$
$$\mathbb{E}\left[\hat{\beta}_{\text{RED}} |\ \beta_1, \sigma_{\text{RED}}, \hat{\beta}_{\text{RED}} > z_\alpha \sigma_{\text{RED}}\right]$$

where OVB, IV and RED denote the naive regression, the IV and the reduced form respectively. β_1 represents the true effect of interest, $\hat{\beta}_{..}$ its estimate, $\sigma_{..}$ the standard error of the estimate and z_α the threshold value of a two-sided test of size α. In all of our analyses, we consider tests of size 5%. One can thus think of z_α as being equal to 1.96.

To compute the exaggeration ratios, we need to know the distribution of the estimates. We first describe a general equation for the exaggeration of a normally distributed estimator. After describing the data generating process, we thus derive the asymptotic distribution of these estimates. We then compute these quantities and analyse how they evolve as a function of the parameters of interests. We then analyse the drivers of the difference between those two quantities. The main parameter of interest is linked the proportion of the variance in the explanatory variable of interest retained for identification.

I. 2 Data generating process

Consider the usual linear regression model to which we add omitted variables. We limit the number of covariates considered to a constant, an observed variable and an omitted variable. For all individual $i \in 1, ..., n$, we make the following assumptions:

$$y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i \tag{1}$$

where $\mathrm{x}_i \in \mathbb{R}^2$ is a vector including a constant and the observed explanatory variable such that $\mathrm{x}_i = (1, x_i)'$, $y_i \in \mathbb{R}$ the observed outcome of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$ an unobserved error term. $\boldsymbol{\beta} \in \mathbb{R}^2$ and $\delta \in \mathbb{R}$ are unknown parameters. $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ and β_1 is the parameter of interest. For readability, we introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$.

We further assume that the usual OLS assumptions hold: $(y_i, \mathrm{x}_i, w_i, u_i)$ are iid, finite second moments, *ie* $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E}[w_i^2] < \infty$, $\mathbb{E}[x_i^2] < \infty$, positive-definiteness of $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']$, u_i is conditional mean-zero, *ie* $\mathbb{E}[u_i | x_i, w_i] = 0$, and is uncorrelated with x_i and w_i, *ie* $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[w_i u_i] = 0$. Note that conditional mean-zero implies that $\mathbb{E}[u_i | x_i] = 0$ (since $\mathbb{E}[u_i | x_i] = \mathbb{E}[\mathbb{E}[u_i | x_i, w_i | x_i]] = \mathbb{E}[0 | x_i] = 0$). We further assume homoskedasticity such that $\mathbb{E}[u_i^2 | x_i, w_i] = \sigma_u^2$ is constant. It implies that $\mathbb{E}[u_i^2 | x_i] = \sigma_u^2$ (since $\mathbb{E}[u_i^2 | x_i] = \mathbb{E}[\mathbb{E}[u_i^2 | x_i, w_i] | x_i] = \mathbb{E}[\sigma_u^2 | x_i] = \sigma_u^2$).

We however assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, we further assume that the unobserved variable is centred, *ie* $\mathbb{E}[w_i] = 0$. We also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^{\perp}$) does not vary with x_i. Thus, $\mathrm{Var}(w_i^{\perp} | x_i) = \mathrm{Var}(w_i^{\perp})$.

We further assume that there exists a valid instrumental variable z_i for x_i. Let $\mathrm{z} = (1, z_i)' \in \mathbb{R}^2$. Note that the model is just-identified. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. To simplify interpretation, we further assume a data generating process for x_i:

$$x_i = \eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i \tag{2}$$

where $(\eta_0, \eta_1, \gamma) \in \mathbb{R}^3$ are unknown parameters. We assume that ϵ_i is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i \epsilon_i] = 0$ and $\mathbb{E}[w_i \epsilon_i] = 0$. Note that, since x_i and w_i are correlated, $\gamma \neq 0$. We also assume homoskedasticity for this term, such that $\mathbb{E}[\epsilon_i^2 | z_i, w_i] = \sigma_\epsilon^2$ is constant. It implies that $\mathbb{E}[\epsilon_i^2 | z_i] = \sigma_\epsilon^2$ (since $\mathbb{E}[\epsilon_i^2 | z_i] = \mathbb{E}[\mathbb{E}[\epsilon_i^2 | z_i, w_i] | z_i] = \mathbb{E}[\sigma_\epsilon^2 | z_i] = \sigma_\epsilon^2$).

The following Directed Acyclic Graph (DAG) sums up the relationships between the variables. It also helps underline the independence, or not, of some variables.

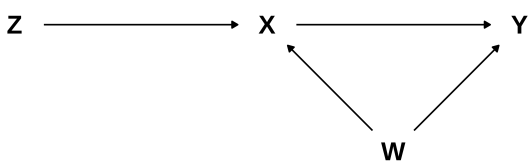

Figure 1 – DAG of the data generating process

In the reduced form case, we consider an analogous DGP to that of the IV. Yet, since we will directly regress y on z, the resulting estimand will be different. To make them comparable, for the reduced form case, in equation 2, we set η_1 to 1. An increase of 1 in z causes to an increase of β_1 in y and the estimands become comparable. Further, we focus on the case where z is an exogenous discrete shock. For clarity, we denote this discrete shock $\mathrm{d}_i = (1, d_i)'$. All other assumptions and features of the DGP remain unchanged. Under these assumptions, plugging equation 2 into equation 1 yields:

$$y_i = \underbrace{\beta_0 + \beta_1\eta_0}_{b_0} + \beta_1 d_i + \underbrace{(\delta + \beta_1\gamma)}_{\kappa} w_i + \underbrace{u_i + \beta_1\epsilon_i}_{\nu_i}$$

The second component of the parameter vector associated with d_i is thus identical to the second component of $\boldsymbol{\beta}$. For the sake of clarity, we also assume that $\eta_0 = 0$ such that $b_0 = \beta_0$ and thus $(b_0, \beta_1) = \boldsymbol{\beta}'$. This assumption is not necessary but clarifies the analysis. Dropping does not change the results for the second component of $\boldsymbol{\beta}$, the component of interest. Thus, in the reduced form case, we have:

$$y_i = \mathrm{d}_i'\boldsymbol{\beta} + \kappa w_i + \nu_i \tag{3}$$

I. 3 Summary of the key assumptions

To ease the use of assumptions in the following sections, we summarize some that we will explicitly mobilize:

— $y_i = \mathrm{x}_i'\boldsymbol{\beta} + \delta w_i + u_i$ ($y_i = \mathrm{d}_i'\boldsymbol{\beta} + \kappa w_i + \nu_i$ in the reduced form case).
— $x_i = \eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i$ (with $x_i = 1 \cdot d_i + \gamma w_i + \epsilon_i$ in the reduced form case).
— $\mathbb{E}[u_i|x_i, w_i] = 0$ and $\mathbb{E}[u_i|x_i] = 0$
— $\mathbb{E}[u_i^2|x_i, w_i] = \sigma_u^2$ and $\mathbb{E}[u_i^2|x_i] = \sigma_u^2$
— $\mathbb{E}[w_i] = 0$
— $\mathbb{E}[z_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$
— $\mathbb{E}[z_i \epsilon_i] = 0$ and $\mathbb{E}[w_i \epsilon_i] = 0$
— $\mathbb{E}[\epsilon_i^2|z_i, w_i] = \sigma_\epsilon^2$ and $\mathbb{E}[\epsilon_i^2|z_i] = \sigma_\epsilon^2$
— $\mathrm{Var}(w_i^\perp|x_i) = \mathrm{Var}(w_i^\perp)$

I. 4 The models to estimate

We consider 3 models:

1. A "naive" regression of y on x (with w omitted). Since, we do not observe w, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + v \tag{4}$$

   where $v = (v_1, ..., v_n)' \in \mathbb{R}^n$ and by definition of the projection, $\mathbb{E}[X'v] = 0$. Note that we use capital letters to denote matrices (for instance $X = (\mathrm{x}_1', ..., \mathrm{x}_n')'$, with $\forall i, \mathrm{x}_i' = (1, x_i)$).

2. An IV regression in which we regress y on x instrumented by z. More precisely, we instrument $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We thus consider a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$.

3. A reduced form equation in which we regress y on the exogenous shock D:

$$y = D\boldsymbol{\beta}_{\text{RED}} + e$$

where $e = (e_1, ..., e_n)' \in \mathbb{R}^n$ and by the exogeneity assumption $\mathbb{E}[D'e] = 0$.

II Exaggeration ratio and its properties

We first compute the exaggeration ratio for a normally distributed estimator and study its general properties. This then enables us to illustrate the intuition behind the trade-off discussed in this paper.

II. 1 A general expression for the exaggeration ratio

Lemma 1. *For an estimator $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$ of a true effect of magnitude β_1 and such that $\beta = \beta_1 + b$, the exaggeration ratio E defined as the ratio expected value of the estimate conditional on the estimate being statistically significant over the true effect is:*

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\mathbb{E}\left[\hat{\beta} |\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]}{\beta_1} = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)$$

Note that Gelman & Carlin (2014) consider the expected value of the *absolute value* of the estimate: $\mathbb{E}\left[|\hat{\beta}|| \ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right] / \beta_1$. To simplify the derivation, we abstract from the absolute value. Our exaggeration ratio therefore depends on the well-known expected value of a truncated normal distribution instead of on the expected value of a truncated *folded* normal distribution. To avoid using the two terms interchangeably, in this document, we refer to E as the "exaggeration ratio" and $\mathbb{E}\left[\hat{\beta} |\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]$ as "exaggeration".

Proof. Since $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$, its support is $\mathbb{R}$ and pdf is $f_{\hat{\beta}} : x \mapsto \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\beta)^2}{2\sigma^2}}$. Thus, by definition of the conditional expectation,

$$\mathbb{E}\left[\hat{\beta} |\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right] = \frac{1}{\Pr(\hat{\beta} > z_\alpha \sigma)} \int_{z_\alpha \sigma}^{\infty} x \times f_{\hat{\beta}}(x) \mathrm{d}x$$

Now,

$$\begin{aligned}
\Pr(\hat{\beta} > z_\alpha \sigma) &= \int_{z_\alpha \sigma}^{\infty} f_{\hat{\beta}}(x) \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha \sigma}^{\infty} e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t && \text{by the change of variable } x = \sigma t + \beta \\
&= 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) && \text{with } \Phi \text{ the cdf of the std. normal} \\
&= \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) && \text{since } \forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)
\end{aligned}$$

and,

$$\int_{z_\alpha\sigma}^{\infty} x \times f_{\hat{\beta}}(x)\mathrm{d}x = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha\sigma}^{\infty} x e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x$$

$$= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} (\sigma t + \beta) e^{-\frac{t^2}{2}} \mathrm{d}t \qquad \text{by the same change of variable}$$

$$= \beta \times \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t + \sigma \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} t e^{-\frac{t^2}{2}} \mathrm{d}t$$

$$= \beta \left(1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right)\right) + \sigma \frac{1}{\sqrt{2\pi}} \left[-e^{-\frac{t^2}{2}}\right]_{z_\alpha - \frac{\beta}{\sigma}}^{\infty}$$

$$= \beta\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \frac{1}{\sqrt{2\pi}} e^{-\frac{(z_\alpha - \beta)^2}{2\sigma^2}}$$

$$= \beta\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma\phi\left(z_\alpha - \frac{\beta}{\sigma}\right) \qquad \text{with } \phi \text{ the pdf of the std. norm.}$$

$$= \beta\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma\phi\left(\frac{\beta}{\sigma} - z_\alpha\right) \qquad \text{since } \forall x \in \mathbb{R},\ \phi(-x) = \phi(x)$$

Thus, we have,

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\beta}{\beta_1} + \frac{\sigma}{\beta_1} \frac{\phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}{\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}$$

where ϕ and Φ are respectively the pdf and the cdf of the standard normal distribution. We then introduce the inverse Mills ratio, the function $\lambda : x \mapsto \phi(x)/\Phi(x)$. We also consider an estimator whose mean can be decomposed in $\beta = \beta_1 + b$, with β_1 the true effect and b a bias independent of β_1. We can then write the exaggeration ratio as:

$$\boxed{E(\beta, \sigma, \beta_1, z_\alpha) = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)} \tag{5}$$

□

Note that since this document is a thorough description of the analysis, we chose to derive this expression analytically. We could have obtained it directly using properties of a truncated normal or directly using the expression of the inverse Mills ratio.

Another important metric in this paper is statistical power. It defined as the probability of rejecting the null of no effect when it is actually true. Hence, for a usual two-sided t-test with $n \gg k$, with k

the number of covariates, the statistical power p is given by:

$$\begin{aligned} p(\beta, \sigma, \beta_1, z_\alpha) &= \Pr(\hat{\beta} < -z_\alpha \sigma) + \Pr(\hat{\beta} > z_\alpha \sigma) \\ &= \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} < -z_\alpha - \frac{\beta}{\sigma}\right) + \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} > z_\alpha - \frac{\beta}{\sigma}\right) \\ &= \Phi\left(-z_\alpha - \frac{\beta}{\sigma}\right) + 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) \qquad \text{since } \frac{\hat{\beta} - \beta}{\sigma} \sim \mathcal{N}(0,1) \end{aligned}$$

And thus, since $\forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)$,

$$\boxed{p(\beta, \sigma, \beta_1, z_\alpha) = 1 - \Phi\left(\frac{\beta}{\sigma} + z_\alpha\right) + \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)} \tag{6}$$

II. 2 Evolution of the exaggeration ratio with σ, β and b

For simplicity, we focus on the case of $\beta_1 > 0$ and $b > 0$. The inverse Mills ratio is a decreasing function. It thus easily follows that E increases with σ, the standard error of the estimator. The more precise the estimator, the less exaggeration. In the case of an unbiased estimator or of , it also easily follows that E is a decreasing function of β_1, the true effect size. These two results have been proven by Lu et al. (2019) and Zwet & Cator (2021) in the case of an exaggeration defined as the expected value of the absolute value of the estimate, as in Gelman & Carlin (2014). Figure 2 gives an intuition for the relationship between the precision of an estimator and exaggeration by ploting the distribution of estimates for two unbiased estimators with identical mean but different variances and plot their distributions, distinguishing between estimates that are and are not significant.

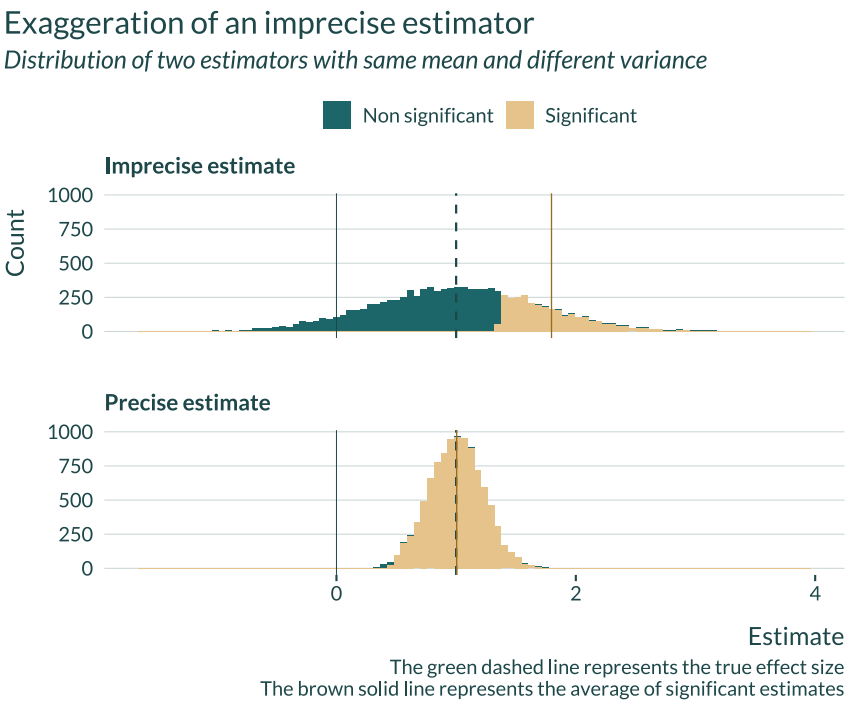

Figure 2 – Intuition for exaggeration of imprecise estimates

In the case of the precise estimator, the mean of statistically significant estimates is almost equal to the true effect. The estimator is unbiased and almost all estimates are significant. However, in the case of the imprecise estimator, only a sub-sample of the estimates are significant: those that are far away enough from 0 (at least 1.96 s.d.). As a consequence, the mean of statistically significant estimates is different from the true effect, even if the estimator is unbiased. This hints to the trade-off between biased estimates from naive regressions and less precise causal inference estimates.

III Asymptotic distribution of the estimates

As discussed in section I. 1, the goal is to compare the exaggeration ratios in the case of the naive regression to that of the corresponding reduced form and IV. As discussed in section II, in order to compute the exaggeration ratios for the various estimators, we need to know their distributions. In this section, we thus **derive the asymptotic distributions of the various estimators of interest**.

III. 1 Naive regression (OVB)

Lemma 2. *Based on the data generating process described in equation 1, for the naive OLS estimate $\hat{\beta}_{\text{OVB}}$ of the regression of y on x, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{OVB}} - \left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{\sigma_x^2}\right)$$

Proof. We first compute the bias of the estimator. From equation 4 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'v \\
\Rightarrow\ & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'v]}_{0} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'y] \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] && \text{cf eq. 1} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta + \underbrace{\mathbb{E}[X'u]}_{0} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned}$$

Since X and w are correlated and $\delta \neq 0$, $\boxed{\boldsymbol{\beta}_{\text{OVB}} \neq \boldsymbol{\beta}}$. $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient matrix of the projection of w on X. A similar derivation and discussion is carried out in Hansen (2022) for instance.

Let $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ denote the least square estimator of $\boldsymbol{\beta}_{\text{OVB}}$. We thus have:

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'y \tag{7}$$

$$\Rightarrow \quad \hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'(X\boldsymbol{\beta}_{\text{OVB}} + v) \qquad \text{cf eq. 4}$$

$$\Rightarrow \quad \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X] = \boldsymbol{\beta}_{\text{OVB}} + (X'X)^{-1}\underbrace{\mathbb{E}[X'v]}_{0}$$

$$\Rightarrow \quad \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X]] = \boldsymbol{\beta}_{\text{OVB}} \qquad \text{by the Law of Iterated Expect. (LIE)}$$

Thus, $\underline{\hat{\boldsymbol{\beta}}_{\text{OVB}} \text{ is an unbiased estimator of } \boldsymbol{\beta}_{\text{OVB}} \text{ but a biased estimator of } \boldsymbol{\beta}}$. We then compute its asymptotic distribution.

We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right)$$

By the Weak Law of Large Numbers (WLLN), $\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Since $\mathbb{E}[\mathrm{x}_i v_i] = 0$, by the Central Limit Theorem (CLT), $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2])$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta} + \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}/n\right)} \tag{8}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x \\ x & x^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \eta_1 \underbrace{\mathrm{cov}(z_i, w_i)}_{0} + \gamma \underbrace{\mathrm{var}(w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{9}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[v_i^2|\mathrm{x}_i]$, noting that:

$$\begin{aligned}
v_i &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i'\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta\underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^{\perp} \qquad\qquad \text{where } w_i^{\perp} = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[v_i^2|\mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^{\perp})^2|x_i] \\
&= \mathbb{E}[u_i^2|x_i] + 2\delta\mathbb{E}[u_i w_i^{\perp}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + 2\delta\left(\mathbb{E}[u_i w_i|x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i|x_i]}_{0}\right) + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i\underbrace{\mathbb{E}[u_i|x_i, w_i]}_{0}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^{\perp})^2|x_i] = \mathrm{Var}(w_i^{\perp}|x_i) + \mathbb{E}[w_i^{\perp}|x_i]^2$. Now, since $w_i^{\perp}$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^{\perp}|x_i] = 0$. And thus, since by assumption $\mathrm{Var}(w_i^{\perp}|x_i) = \mathrm{Var}(w_i^{\perp})$,

$$\begin{aligned}
\mathbb{E}[(w_i^{\perp})^2|x_i] &= \mathrm{Var}(w_i^{\perp}|x_i) \\
&= \mathrm{Var}(w_i^{\perp}) \\
&= \mathbb{E}[(w_i^{\perp})^2] - \mathbb{E}[w_i^{\perp}]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x\underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 - 2\frac{\gamma^2\sigma_w^4}{\sigma_x^2} + \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2 - \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2\sigma_w^2$. Under this condition,

$$\mathbb{E}[v_i^2|\mathrm{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{10}$$

Thus, under our set of assumptions, $\mathbb{E}[v_i^2|\mathrm{x}_i]$ does not depend on x_i and $\mathbb{E}[v_i^2|\mathrm{x}_i] = \mathbb{E}[v_i^2]$. We denote this quantity σ_v^2.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, noting that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'v_i^2] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\sigma_v^2] = \sigma_v^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$. And thus $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \sigma_v^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Plugin this and equation 9 into equation 8, we get, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$:

$$\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_v^2}{n\ \sigma_x^2}\right)$$

And substituting for σ_v^2,

$$\boxed{\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}\right)} \tag{11}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{OVB}} &= \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \\ \sigma_{\text{OVB}}^2 &= \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2} \end{aligned} \tag{12}$$

III. 2 Instrumental Variable

Lemma 3. *Based on the sus-mentioned data generating process, for the IV estimate $\hat{\beta}_{\text{IV}}$ of the regression of y on x instrumented by z, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{\eta_1^2\sigma_z^2}\right)$$

Proof. We first derive the bias of the estimator.

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{\text{IV}} &= (Z'X)^{-1}Z'y \\ &= (Z'X)^{-1}Z'(X\boldsymbol{\beta} + \delta w + u) \\ &= \boldsymbol{\beta} + (Z'X)^{-1}Z'w\delta + (Z'X)^{-1}Z'u \end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}}|X,Z] = \boldsymbol{\beta} + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'w]}_{0}\delta + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'u]}_{0} = \boldsymbol{\beta}$$

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i'\right)^{-1}\delta\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i'\right)^{-1}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i u_i\right)$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']$ and since $\mathbb{E}[\mathrm{z}_i w_i] = 0$ and $\mathbb{E}[\mathrm{z}_i u_i] = 0$, the Central Limit Theorem (CLT) gives $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i w_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i\mathrm{z}_i' w_i^2])$ and $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i u_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i\mathrm{z}_i' u_i^2])$.

Now, $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i' u_i^2] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[u_i^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{z}_i)] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[\sigma_u^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\sigma_u^2] = \sigma_u^2\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$. Now, since z_i is a valid instrument, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{z}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i' w_i^2] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[w_i^2|\mathrm{z}_i]] = \sigma_w^2\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\delta^2\sigma_w^2 + \sigma_u^2)\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)'\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n}\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)'\right)}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i]$, $\mathbb{E}[\mathrm{x}_i\mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \mathbb{E}\begin{bmatrix}1 & z\\ z & z^2\end{bmatrix} = \begin{bmatrix}1 & \mu_z\\ \mu_z & \sigma_z^2 + \mu_z^2\end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i'] = \begin{bmatrix}1 & \mathbb{E}[x_i]\\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i]\end{bmatrix} = \begin{bmatrix}1 & \eta_0 + \eta_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[\epsilon_i]}_{0}\\ \mu_z & \eta_0\mathbb{E}[z_i] + \eta_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i\epsilon_i]}_{0}\end{bmatrix} = \begin{bmatrix}1 & \eta_0 + \eta_1\mu_z\\ \mu_z & \eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2)\end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1} = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix}\eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) & -\eta_0 - \eta_1\mu_z\\ -\mu_z & 1\end{bmatrix}$$

$$\text{and} \quad \left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix}\eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) & -\mu_z\\ -\eta_0 - \eta_1\mu_z & 1\end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix}\eta_1\sigma_z^2 & -\eta_0\sigma_z^2\\ 0 & \sigma_z^2\end{bmatrix} = \begin{bmatrix}1 & -\frac{\eta_0}{\eta_1}\\ 0 & \frac{1}{\eta}\end{bmatrix}$$

And finally,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix} 2\eta_0\mu_z + \eta_1(\sigma_z^2+\mu_z^2) + \frac{\eta_0^2}{\eta_1} & -\mu_z - \frac{\eta_0}{\eta_1} \\ -\mu_z - \frac{\eta_0}{\eta_1} & \frac{1}{\eta_1} \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\boxed{\hat{\beta}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\beta_1,\ \frac{\delta^2\sigma_w^2+\sigma_u^2}{n\ \eta_1^2\sigma_z^2}\right)} \tag{13}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{IV}} &= 0 \\ \sigma_{\text{IV}}^2 &= \frac{\delta^2\sigma_w^2+\sigma_u^2}{n\ \eta_1^2\sigma_z^2} \end{aligned} \tag{14}$$

Note that, since z_i is uncorrelated with w_i and ϵ_i,

$$\begin{aligned} \sigma_x^2 &= \mathrm{var}(x_i) = \mathrm{var}(\eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i) \\ &= \eta_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_\epsilon^2 + 2\eta_1\gamma\underbrace{\mathrm{cov}(z_i, w_i)}_{0} + 2\eta_1\underbrace{\mathrm{cov}(z_i,\epsilon_i)}_{0} + 2\gamma\underbrace{\mathrm{cov}(w_i,\epsilon_i)}_{0} \\ &= \underbrace{\eta_1^2\sigma_z^2}_{V_z^x} + \underbrace{\gamma^2\sigma_w^2+\sigma_\epsilon^2}_{V_{\text{non-}z}^x} \end{aligned}$$

For a given variance of x, <u>the larger the proportion of this variance that depends on z, V_z^x, (and the smaller the non-z component $V_{\text{non-}z}^x$), the more precise the estimate</u>.

Also note that,

$$\begin{aligned} & \mathrm{cov}(x_i, z_i) = \mathrm{cov}(\eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i, z_i) = \eta_1\sigma_z^2 \\ \Rightarrow\ & \rho_{xz} = \mathrm{corr}(x_i, z_i) = \frac{\mathrm{cov}(x_i, z_i)}{\sigma_x\sigma_z} = \eta_1\frac{\sigma_z}{\sigma_x} \\ \Rightarrow\ & \eta_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2 \end{aligned}$$

One can therefore rewrite the formula for the variance of our IV estimator as

$$\sigma_{\text{IV}}^2 = \frac{\delta^2\sigma_w^2+\sigma_u^2}{n\ \sigma_x^2\rho_{xz}^2}$$

This variance formula enables to retrieve the usual result regarding the determinant of the 2SLS variance as discussed in Hansen (2022) for instance. The variance of the estimator increases when σ_u^2, the variance of the error, increases. It decreases with ρ_{xz}, the correlation between x and z, and with σ_x^2, the variance of x, assuming the omitted variable fixed.

III. 3 Reduced form

Lemma 4. *Based on the sus-mentioned data generating process, for the reduced form estimate $\hat{\beta}_{\text{RED}}$ of the regression of y on d, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{RED}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n \times p_1(1-p_1)}\right)$$

where p_1 is the proportion of 1 in d.

Proof. We first derive the bias of the estimator.

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{RED}} &= (D'D)^{-1}D'y \\
&= (D'D)^{-1}D'(D\boldsymbol{\beta} + \kappa w + \nu) \\
&= \boldsymbol{\beta} + \kappa(D'D)^{-1}D'w + (D'D)^{-1}D'\nu
\end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D] = \boldsymbol{\beta} + (D'D)^{-1}\underbrace{\mathbb{E}[D'w]}_{0}\kappa + (D'D)^{-1}\underbrace{\mathbb{E}[D'\nu]}_{\substack{\beta_1\mathbb{E}[D'\epsilon]+\mathbb{E}[D'u] \\ = \beta_1 0 + 0 \ = \ 0}} = \boldsymbol{\beta}$$

By the Law of Iterated Expectations, we get that $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D]] = \beta$ and thus $\hat{\boldsymbol{\beta}}_{\text{RED}}$ is an unbiased estimator of $\boldsymbol{\beta}$. We then derive its asymptotic distribution, noting that:

$$\begin{aligned}
\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) &= \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\kappa\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\nu_i\right) \\
&= \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\kappa\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) \\
&\quad + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\left[\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i u_i\right) + \beta_1\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\epsilon_i\right)\right]
\end{aligned}$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i' \xrightarrow{p} \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$ and since $\mathbb{E}[\mathrm{d}_i w_i] = 0$,$\mathbb{E}[\mathrm{d}_i u_i] = 0$ and $\mathbb{E}[\mathrm{d}_i\epsilon_i] = 0$, the Central Limit Theorem (CLT) gives:

$$\begin{aligned}
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2]) \\
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i u_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2]) \\
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\epsilon_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\epsilon_i^2])
\end{aligned}$$

As for the IV, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[u_i^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{d}_i)] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\sigma_u^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_u^2] = \sigma_u^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. In addition, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\epsilon_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\epsilon_i^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_\epsilon^2] = \sigma_\epsilon^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. Now, since d_i is exogenous, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{d}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[w_i^2|\mathrm{d}_i]] = \sigma_w^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\kappa^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2)\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

Now, since $\kappa = \delta + \beta\gamma$,

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, \{(\delta + \beta\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2\}\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

Notice that, as expected, the term factored in front of $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}$ corresponds to the part of the variance in y that is orthogonal to d, *i.e.*, the variance of the error term of the regression of y on d:

$$\begin{aligned}
\mathrm{Var}\left(y_i^{\perp d}\right) &= \mathrm{Var}\left((\delta + \beta_1\gamma)w_i + u_i + \beta_1\epsilon_i\right) \\
&= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2 \\
&\quad + 2(\delta + \beta_1\gamma)\underbrace{\mathrm{cov}(w_i, u_i)}_{0} + 2\beta_1\underbrace{\mathrm{cov}(u_i, \epsilon_i)}_{0} + 2(\delta + \beta_1\gamma)\beta_1\underbrace{\mathrm{cov}(w_i, \epsilon_i)}_{0} \\
&= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2
\end{aligned}$$

The previous relation can also be written as,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n}\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}$.

$$\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'] = \mathbb{E}\begin{bmatrix} 1 & d \\ d & d^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_d \\ \mu_d & \sigma_d^2 + \mu_d^2 \end{bmatrix}$$

Since d is a vector of dummies, denoting p_1 the proportion of 1 in this vector, $\sigma_d^2 = p_1(1 - p_1)$ and $\mu_d = p_1$. Thus, $\sigma_d^2 + \mu_d^2 = p_1$. As a consequence,

$$\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'] = \begin{bmatrix} 1 & p_1 \\ & \\ p_1 & p_1 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1} = \frac{1}{p_1(1 - p_1)}\begin{bmatrix} p_1 & -p_1 \\ -p_1 & 1 \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{RED}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$, we have:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n \times p_1(1 - p_1)}\right)}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{RED}} &= 0 \\ \sigma^2_{\text{RED}} &= \frac{(\delta+\beta_1\gamma)^2\sigma_w^2+\sigma_u^2+\beta_1^2\sigma_\epsilon^2}{n\times p_1(1-p_1)} \end{aligned} \tag{15}$$

III. 4 Comparison of efficiencies of the IV and naive regression

The IV and naive regression variances are rather similar. Note that $\sigma_x^2\rho_{xz}^2 < \sigma_x^2$ and,

$$\frac{\gamma^2\sigma_w^2}{\sigma_x^2} > 0 \quad \Leftrightarrow \quad 1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2} < 1 \quad \Leftrightarrow \quad \sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) < \sigma_u^2+\delta^2\sigma_w^2$$

Both these inequalities imply that $\boxed{\sigma^2_{\text{IV}} > \sigma^2_{\text{OVB}}}$. While unbiased, the IV estimator is less precise than the naive estimator. This is a well known and intuitive result that we recovered in a particular setting, tied to the modelling assumptions we made. This result is due to the fact that the IV only uses part of the variation in the endogenous dependent variable, the part that is explained by the instrument.

IV Comparison of exaggeration ratios

Now that we have computed the asymptotic distributions of the various estimators, we can compute their exaggeration ratios. We study how they evolve with the values of model parameters and prove the existence of the exaggeration/confounder trade-off in our specific case.

IV. 1 Evolution of the exaggeration ratio with the value of model parameters

We focus on the OVB and IV estimators whose distributions we derived in the previous section in order to understand how their respective exaggeration ratios, E_{OVB} and E_{IV}, evolve with the values of the model parameters described in section I. 2. In particular, we want to understand how exaggeration evolves with the correlation between x and z, ρ_{xz} and the intensity of the OVB, *ie*, δ and γ.

IV. 1.1 Omitted Variable

Plugging in the values for β_{OVB}, σ_{OVB} and b_{OVB}, from equation 12 into equation 5, we get:

$$\begin{aligned} E_{\text{OVB}} &= 1+\frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2}+\frac{\sqrt{\dfrac{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}}{\beta_1}\lambda\left(\frac{\beta_1+\dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}}{\sqrt{\dfrac{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}}-z_\alpha\right) \\ &= 1+\frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2}+\frac{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1+\dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}-z_\alpha\right) \end{aligned}$$

Keeping in mind that λ is a decreasing function, we notice that E_{ovb} decreases with β_1, σ_x and n and increases with σ_u. Analytically deriving the evolution of this quantity with the omitted variable bias, *ie*, with δ, γ and σ_w is more challenging. The direction of this evolution is likely to depend on the value of the parameters. Since this is not the focus of the present paper, we do not dive into this. We however expect that, if the bias is upward (δ and γ of the same sign), we expect the exaggeration to increase with δ, γ and σ_w^2, for realistic values of the various parameters since the usual omitted variable bias $\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)$ is a major component of the exaggeration.

IV. 1.2 Instrumental Variable

Proposition 1. *E_{IV}, the exaggeration ratio of the instrumental variable estimator, decreases with the strength of the instrument η_1 (or equivalently with the correlation between the endegoneous variable x and the instrument z).*

Proof. Plugging in the values for β_{IV}, σ_{IV} and b_{IV}, from equation 14 into equation 5, we get:

$$\begin{aligned} E_{\text{IV}} &= 1 + \frac{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \eta_1^2\sigma_z^2}}}{\beta_1}\lambda\left(\frac{\beta_1}{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \eta_1^2\sigma_z^2}}} - z_\alpha\right) \\ &= 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1|\eta_1|\sigma_z\sqrt{n}}\lambda\left(\frac{\beta_1|\eta_1|\sigma_z\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right) \end{aligned}$$

Since there is no bias, the only factor varying with the values of the model parameters is the standard error of the estimator. Since the exaggeration ratio increases with this standard error, any model parameter positively associated with the later will increase the former. Hence, the exaggeration ratio of the IV estimator, E_{IV}, increases with σ_u, σ_w and δ and decreases with σ_z, η_1 and n. It also decreases with β_1. We thus showed that <u>exaggeration decreases with the strength of the instrument</u>. Now, note that:

$$\begin{aligned} \sigma_y^2 &= \text{var}(y_i) = \text{var}(\beta_0 + \beta_1 x_i + \delta w_i + u_i) \\ &= \beta_1^2\sigma_x^2 + \delta^2\sigma_w^2 + \sigma_u^2 + 2\beta_1\delta\underbrace{\text{cov}(x_i, w_i)}_{\gamma\sigma_w^2} + 2\beta_1\underbrace{\text{cov}(x_i, u_i)}_{0} + 2\delta\underbrace{\text{cov}(w_i, u_i)}_{0} \\ &= \beta_1^2\sigma_x^2 + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\ &= \beta_1^2(\eta_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_\epsilon^2) + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\ &= \underbrace{\beta_1^2\eta_1^2\sigma_z^2}_{V_z^y} + \underbrace{(\beta_1\gamma + \delta)^2\sigma_w^2 + (\beta_1^2\sigma_\epsilon^2 + \sigma_u^2)}_{V_{\text{non-}z}^y} \end{aligned}$$

The larger the parameters that are positively associated with E_{IV} (σ_u, σ_w and δ), the larger the proportion of the variance of y that does not depend on z, $V_{\text{non-}z}^y$. Hence, for a given variance of y, the smaller the proportion of this variance that depends on z, V_z^y, (and the larger the non-z component $V_{\text{non-}z}^y$), the less precise the estimate and the larger the exaggeration. <u>The smaller the proportion of the variance in y that depends on z, the larger the exaggeration of the IV estimator</u>.

Since $\eta_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, we can also express this exaggeration ratio as:

$$E_{\text{IV}} = 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right)$$

Note that E_{IV} decreases with ρ_{xz}, the correlation between the endogenous variable x and the instrument z.

□

IV. 2 The exaggeration/confounders trade-off

The objective of this paper is to compare "biases" resulting from exaggeration and confounders. In this document, we thus aim to compare E_{OVB} to E_{IV}. We thus define $\Delta = E_{\text{IV}} - E_{\text{OVB}}$ and analyse whether and for which parameters values $\Delta > 0$. In particular, we focus on the strength of the IV (or the correlation between the instrument and the endogenous variable).

Proposition 2. *Under the data generating process described in equation 1, for some parameter values,* $E_{\text{IV}} > E_{\text{OVB}}$.

Proof. Combining lemmas 2, 3 and 1, we get:

$$\begin{aligned}\Delta = E_{\text{IV}} - E_{\text{OVB}} = \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right) - \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} \\ - \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right)\end{aligned}$$

First, note that only E_{IV} depends on ρ_{xz}. Since E_{IV} decreases with ρ_{xz}, Δ also decreases with ρ_{xz}. Now, for fixed and finite values of the other parameters, and since we assumed that $\beta_1 > 0$,

$$\begin{aligned}\lim_{\rho_{xz}\to 0^+}\frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}} &= +\infty \\ \lim_{\rho_{xz}\to 0^+}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right) &= \lambda(-z_\alpha) \in \mathbb{R}_+\end{aligned}$$

Since the other components of Δ are finite and do not depend on ρ_{xz}, $\lim_{\rho_{xz}\to 0^+}\Delta = +\infty$. Since Δ is a continuous function of ρ_{xz} on $(0, 1]$, for any set of parameter values consistent with our hypotheses, $\exists\rho_{xz} \in (0, 1]$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$. Since $\eta_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, it is also straigthforward to show, for any set of parameter values consistent with our hypotheses, $\exists\eta_1 \in \mathbb{R}_+^*$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$.

□

We thus have shown that, if the correlation between the instrument and the endogenous variable is low, the exaggeration bias of the IV can be larger than that of the "naive" regression, even in the

presence of confounders. The trade-off, *ie* the value of Δ, is affected by this correlation, or the strength of the instrument.

Yet, the value and the sign of Δ depend on the value of the other parameters in a non-trivial maner. In a concrete setting, it might therefore be difficult to identify how "low" the correlation has to be for Δ to be positive.

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